

Normal-Mode Splitting Functions and Earth-Model Sensitivities

A Gravity-Free Normal-Mode Analogue of the Liu–Tromp Kernel Framework

Abstract

These notes derive the mathematical connection between a laterally heterogeneous Earth model and normal-mode splitting functions. The derivation is deliberately operator-first: the Earth model perturbs an elastodynamic pencil, the perturbation is restricted to a degenerate normal-mode eigenspace, and the resulting finite-dimensional splitting matrix is expanded in spherical harmonics.

The bulk part of the derivation is the standard first-order normal-mode perturbation formula. The interface/topography part is written in the same structural form as Liu & Tromp’s finite-frequency adjoint kernels: a volumetric term, a scalar boundary-topography term, and, on fluid–solid boundaries, an additional surface-gradient boundary term. Thus interface topography is *not* treated as a naive sampled bulk-energy density. It is treated as a true shape derivative of the elastic/acoustic transmission problem.

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1 Purpose and scope

The aim is to connect three objects:

- (i) a perturbation of an Earth model,

$$\delta m = (\delta\lambda, \delta\mu, \delta\rho, \{\eta_\alpha\}),$$

where η_α denotes normal topography of an internal boundary;

- (ii) a finite-dimensional splitting matrix for a degenerate normal-mode multiplet,

$$H_{mm'}^{(n,\ell)}, \quad -\ell \leq m, m' \leq \ell;$$

- (iii) splitting coefficients and splitting functions,

$$c_{st}^{(2)}(n, \ell), \quad F_{n\ell}^{(2)}(\theta, \phi) = \sum_{s,t} c_{st}^{(2)}(n, \ell) Y_s^t(\theta, \phi).$$

The superscript (2) reminds us that, unless otherwise stated, these coefficients describe perturbations to ω^2 . Conventional frequency-splitting coefficients are obtained by dividing by $2\omega_{n\ell}$:

$$c_{st}(n, \ell) = \frac{c_{st}^{(2)}(n, \ell)}{2\omega_{n\ell}}. \quad (1.1)$$

The reference problem is an SNREI-like, spherically symmetric, non-rotating, gravity-free elastic/acoustic Earth. The gravity-free assumption is made only to keep the normal-mode algebra transparent. Liu & Tromp treat the more general rotating, self-gravitating problem and then give a no-rotation/no-self-gravity reduction for the kernel structure. These notes use precisely that reduced structure as the guide for the boundary terms.

2 Geometry, interfaces, and jump conventions

Let $\Omega \subset \mathbb{R}^3$ denote the Earth model domain. The collection of interfaces is

$$\Sigma = \partial\Omega \cup \Sigma_{\text{SS}} \cup \Sigma_{\text{FS}},$$

where Σ_{SS} denotes solid–solid interfaces and Σ_{FS} denotes fluid–solid interfaces. For an internal interface, $\hat{\mathbf{n}}$ points from the interior “−” side to the exterior “+” side, and

$$[[q]] := q^+ - q^-.$$

The reference parameters are radial:

$$\rho_0 = \rho_0(r), \quad \lambda_0 = \lambda_0(r), \quad \mu_0 = \mu_0(r),$$

piecewise smooth with jumps at the interfaces. In fluid regions $\mu_0 = 0$ in the isotropic elastic-acoustic idealization.

The stress tensor is

$$\mathbf{T}_0(\mathbf{u}) = \lambda_0(\nabla \cdot \mathbf{u})\mathbf{I} + 2\mu_0\boldsymbol{\varepsilon}(\mathbf{u}), \quad \boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^\top). \quad (2.1)$$

The traction vector is

$$\mathbf{t}_0(\mathbf{u}) := \hat{\mathbf{n}} \cdot \mathbf{T}_0(\mathbf{u}). \quad (2.2)$$

The correct transmission conditions are:

All interfaces. Traction is continuous:

$$\llbracket \hat{\mathbf{n}} \cdot \mathbf{T}_0(\mathbf{u}) \rrbracket = 0 \quad \text{on } \Sigma. \quad (2.3)$$

Solid–solid interfaces. Displacement is continuous:

$$\llbracket \mathbf{u} \rrbracket = 0 \quad \text{on } \Sigma_{\text{SS}}. \quad (2.4)$$

Fluid–solid interfaces. Tangential slip is allowed, but normal displacement is continuous:

$$\llbracket \hat{\mathbf{n}} \cdot \mathbf{u} \rrbracket = 0 \quad \text{on } \Sigma_{\text{FS}}. \quad (2.5)$$

Free surface. The exterior material parameters are taken to be zero, so traction continuity reduces to

$$\hat{\mathbf{n}} \cdot \mathbf{T}_0(\mathbf{u}) = 0 \quad \text{on } \partial\Omega. \quad (2.6)$$

These conditions are important. Treating all internal interfaces as if $\llbracket \mathbf{u} \rrbracket = 0$ is valid for a solid-only toy Earth, but not for an Earth model containing fluid–solid boundaries such as the CMB or ICB.

3 The gravity-free normal-mode eigenproblem

The time-domain gravity-free equation is

$$\rho_0 \partial_t^2 \mathbf{s} - \nabla \cdot \mathbf{T}_0(\mathbf{s}) = 0. \quad (3.1)$$

With $\mathbf{s}(\mathbf{x}, t) = \mathbf{u}(\mathbf{x})e^{-i\omega t}$, this becomes

$$-\nabla \cdot \mathbf{T}_0(\mathbf{u}) = \omega^2 \rho_0 \mathbf{u}. \quad (3.2)$$

Define the elastic operator and mass operator by

$$\mathcal{L}_0 \mathbf{u} := -\nabla \cdot \mathbf{T}_0(\mathbf{u}), \quad \mathbf{M}_0 \mathbf{u} := \rho_0 \mathbf{u}. \quad (3.3)$$

The normal-mode problem is the generalized eigenproblem

$$\mathcal{L}_0 \mathbf{u} = \omega^2 \mathbf{M}_0 \mathbf{u}. \quad (3.4)$$

Let V denote the energy space of admissible displacements satisfying the appropriate trace conditions on Σ_{SS} and Σ_{FS} . The weak forms are

$$a_0(\mathbf{u}, \mathbf{v}) = \int_{\Omega} [\lambda_0 (\nabla \cdot \mathbf{u})^* (\nabla \cdot \mathbf{v}) + 2\mu_0 \boldsymbol{\varepsilon}(\mathbf{u})^* : \boldsymbol{\varepsilon}(\mathbf{v})] \, dV, \quad (3.5)$$

$$m_0(\mathbf{u}, \mathbf{v}) = \int_{\Omega} \rho_0 \mathbf{u}^* \cdot \mathbf{v} \, dV. \quad (3.6)$$

The weak eigenproblem is

$$a_0(\mathbf{u}, \mathbf{v}) = \omega^2 m_0(\mathbf{u}, \mathbf{v}), \quad \forall \mathbf{v} \in V. \quad (3.7)$$

For a spherically symmetric reference Earth, modes are labelled by (n, ℓ, m) :

$$\mathcal{L}_0 \mathbf{u}_{n\ell m} = \omega_{n\ell}^2 \mathbf{M}_0 \mathbf{u}_{n\ell m}, \quad m = -\ell, \dots, \ell. \quad (3.8)$$

The $2\ell + 1$ functions $\{\mathbf{u}_{n\ell m}\}$ span a degenerate eigenspace,

$$\mathcal{E}_{n\ell} = \text{span}\{\mathbf{u}_{n\ell m} : -\ell \leq m \leq \ell\}. \quad (3.9)$$

We use the normalization

$$m_0(\mathbf{u}_{n\ell m}, \mathbf{u}_{n\ell m'}) = \delta_{mm'}. \quad (3.10)$$

4 Hellmann–Feynman formula for the generalized pencil

Let

$$a_\epsilon = a_0 + \epsilon \delta a + o(\epsilon), \quad m_\epsilon = m_0 + \epsilon \delta m + o(\epsilon).$$

For a simple eigenvalue, with $m_0(\mathbf{u}, \mathbf{u}) = 1$,

$$\delta\omega^2 = \delta a(\mathbf{u}, \mathbf{u}) - \omega^2 \delta m(\mathbf{u}, \mathbf{u}). \quad (4.1)$$

For a degenerate multiplet, one first restricts the perturbed pencil to the unperturbed eigenspace. The splitting matrix is

$$\boxed{H_{mm'}^{(n,\ell)} = \delta a(\mathbf{u}_{n\ell m}, \mathbf{u}_{n\ell m'}) - \omega_{n\ell}^2 \delta m(\mathbf{u}_{n\ell m}, \mathbf{u}_{n\ell m'}) + H_{mm'}^\Sigma} \quad (4.2)$$

The eigenvalues of $H^{(n,\ell)}$ are the first-order shifts in ω^2 , and its eigenvectors give the first-order mixtures of the m components.

The term $H_{mm'}^\Sigma$ denotes the contribution from perturbing the location of interfaces. This term is *not* a naive jump in the bulk energy density. It is the boundary part of the shape derivative of the transmission problem, described in Section 6.

5 Bulk model perturbations

Let the bulk parameters be perturbed by

$$\lambda = \lambda_0 + \delta\lambda, \quad \mu = \mu_0 + \delta\mu, \quad \rho = \rho_0 + \delta\rho.$$

For isotropic absolute perturbations, the bulk terms in (4.2) are

$$H_{mm'}^{\text{bulk}} = \int_{\Omega} \left[\delta\lambda S_{mm'}^\lambda + \delta\mu S_{mm'}^\mu + \delta\rho S_{mm'}^\rho \right] dV, \quad (5.1)$$

$$S_{mm'}^\lambda := (\nabla \cdot \mathbf{u}_m)^* (\nabla \cdot \mathbf{u}_{m'}), \quad (5.2)$$

$$S_{mm'}^\mu := 2 \boldsymbol{\varepsilon}(\mathbf{u}_m)^* : \boldsymbol{\varepsilon}(\mathbf{u}_{m'}), \quad (5.3)$$

$$S_{mm'}^\rho := -\omega_{n\ell}^2 \mathbf{u}_m^* \cdot \mathbf{u}_{m'}. \quad (5.4)$$

Here and below, when no ambiguity is possible,

$$\mathbf{u}_m := \mathbf{u}_{n\ell m}.$$

For an anisotropic parameterization, the compact form is

$$H_{mm'}^{\text{bulk}} = \int_{\Omega} \nabla \mathbf{u}_m^* : \delta \mathbf{c} : \nabla \mathbf{u}_{m'} dV - \omega_{n\ell}^2 \int_{\Omega} \delta\rho \mathbf{u}_m^* \cdot \mathbf{u}_{m'} dV. \quad (5.5)$$

This is the frequency-domain normal-mode counterpart of the forward–adjoint kernel product used in finite-frequency tomography.

6 Boundary and interface topography

Let an interface Σ_α be displaced by a small normal topography

$$\eta_\alpha(\theta, \phi).$$

Positive η_α means displacement in the direction of $\hat{\mathbf{n}}$.

The Liu–Tromp kernel structure shows that a topographic perturbation produces boundary terms of the form

$$\text{scalar boundary kernel} \times \eta_\alpha \quad \text{plus, on fluid–solid boundaries,} \quad \text{vector boundary kernel} \cdot \nabla_\Sigma \eta_\alpha. \quad (6.1)$$

Thus the normal-mode shape contribution is written as

$$H_{mm'}^\Sigma = \sum_{\Sigma_\alpha \subset \partial\Omega \cup \Sigma_{SS}} \int_{\Sigma_\alpha} \eta_\alpha \mathcal{B}_{mm'}^\alpha \, dS + \sum_{\Sigma_\alpha \subset \Sigma_{FS}} \int_{\Sigma_\alpha} (\eta_\alpha \mathcal{B}_{mm'}^\alpha + \mathcal{C}_{mm'}^\alpha \cdot \nabla_\Sigma \eta_\alpha) \, dS. \quad (6.2)$$

The scalar and vector boundary kernels are bilinear in the two modes $\mathbf{u}_m, \mathbf{u}_{m'}$. In the gravity-free analogue of the Liu–Tromp convention, define

$$\mathcal{B}_{mm'}^\alpha := -\llbracket \rho_0 \omega_{nl}^2 \mathbf{u}_m^* \cdot \mathbf{u}_{m'} - \nabla \mathbf{u}_m^* : \mathbf{c}_0 : \nabla \mathbf{u}_{m'} + \mathbf{t}_0(\mathbf{u}_{m'}) \cdot \partial_n \mathbf{u}_m^* + \mathbf{t}_0(\mathbf{u}_m)^* \cdot \partial_n \mathbf{u}_{m'} \rrbracket, \quad (6.3)$$

$$\mathcal{C}_{mm'}^\alpha := \llbracket (\hat{\mathbf{n}} \cdot \mathbf{T}_0(\mathbf{u}_m)^* \cdot \hat{\mathbf{n}}) \mathbf{u}_{m'} + (\hat{\mathbf{n}} \cdot \mathbf{T}_0(\mathbf{u}_{m'}) \cdot \hat{\mathbf{n}}) \mathbf{u}_m^* \rrbracket. \quad (6.4)$$

Here

$$\partial_n := \hat{\mathbf{n}} \cdot \nabla.$$

The vector kernel $\mathcal{C}_{mm'}^\alpha$ contributes only on fluid–solid boundaries. On a solid–solid boundary the corresponding surface-gradient term vanishes because the full displacement is continuous across the interface.

Important correction. The topography kernel is not

$$\llbracket \lambda_0 S^\lambda + \mu_0 S^\mu + \rho_0 S^\rho \rrbracket.$$

That expression is only the naive derivative of a domain integral with the mode field held fixed. It misses the boundary terms produced by differentiating the transmission conditions. The correct shape derivative contains normal-derivative and traction terms, and on fluid–solid boundaries it also contains the surface-gradient term in (6.2).

Equivalent scalar form on closed fluid–solid interfaces. If Σ_α is closed, the surface-gradient term can be integrated by parts:

$$\int_{\Sigma_\alpha} \mathcal{C}_{mm'}^\alpha \cdot \nabla_\Sigma \eta_\alpha \, dS = - \int_{\Sigma_\alpha} (\nabla_\Sigma \cdot \mathcal{C}_{mm'}^\alpha) \eta_\alpha \, dS. \quad (6.5)$$

Thus, on a closed fluid–solid interface,

$$\int_{\Sigma_\alpha} (\eta_\alpha \mathcal{B}_{mm'}^\alpha + \mathcal{C}_{mm'}^\alpha \cdot \nabla_\Sigma \eta_\alpha) \, dS = \int_{\Sigma_\alpha} \eta_\alpha (\mathcal{B}_{mm'}^\alpha - \nabla_\Sigma \cdot \mathcal{C}_{mm'}^\alpha) \, dS. \quad (6.6)$$

This is the normal-mode analogue of the effective boundary kernel

$$K_d^{\text{eff}} = K_d - \nabla_\Sigma \cdot \mathbf{K}_d$$

used for fluid–solid boundary topography in the finite-frequency formulation.

7 Spherical-harmonic expansion of bulk perturbations

Expand the bulk perturbations in spherical harmonics:

$$\delta m_p(r, \theta, \phi) = \sum_{s=0}^{\infty} \sum_{t=-s}^s \delta m_{st}^p(r) Y_s^t(\theta, \phi), \quad p \in \{\lambda, \mu, \rho\}. \quad (7.1)$$

Similarly expand interface topographies:

$$\eta_\alpha(\theta, \phi) = \sum_{s=0}^{\infty} \sum_{t=-s}^s \eta_{\alpha;st} Y_s^t(\theta, \phi). \quad (7.2)$$

For a spherically symmetric reference Earth, angular momentum algebra separates the splitting matrix into fixed angular coupling coefficients and radial kernels. The basic scalar coupling coefficient is the Gaunt integral

$$\mathcal{Q}_{mm';st}^{(\ell)} := \int_{S^2} Y_\ell^{m*} Y_s^t Y_\ell^{m'} d\Omega. \quad (7.3)$$

With Condon–Shortley normalized complex spherical harmonics,

$$\mathcal{Q}_{mm';st}^{(\ell)} = (-1)^m \sqrt{\frac{(2\ell+1)^2(2s+1)}{4\pi}} \begin{pmatrix} \ell & s & \ell \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \ell & s & \ell \\ -m & t & m' \end{pmatrix}. \quad (7.4)$$

The selection rules are:

$$0 \leq s \leq 2\ell, \quad (7.5)$$

$$-m + t + m' = 0, \quad (7.6)$$

$$2\ell + s \text{ even}. \quad (7.7)$$

Thus scalar self-coupling within a single (n, ℓ) multiplet sees only even $s \leq 2\ell$.

8 Radial kernels for bulk splitting

Define the radial bulk kernels by the angular projections

$$\int_{S^2} S_{mm'}^p(r, \theta, \phi) Y_s^t(\theta, \phi) d\Omega = \mathcal{Q}_{mm';st}^{(\ell)} K_s^p(r; n, \ell), \quad (8.1)$$

for $p \in \{\lambda, \mu, \rho\}$. This definition is safer than imposing a pointwise reduction of the strain density. The strain term generally contains tensor-spherical-harmonic contractions; only after angular integration does it reduce to Gaunt coefficients and radial factors.

With (7.1) and (8.1), the bulk splitting matrix becomes

$$H_{mm'}^{\text{bulk}} = \sum_{s,t} \mathcal{Q}_{mm';st}^{(\ell)} c_{\text{bulk};st}^{(2)}(n, \ell), \quad (8.2)$$

where

$$c_{\text{bulk};st}^{(2)}(n, \ell) = \sum_{p \in \{\lambda, \mu, \rho\}} \int_0^a r^2 K_s^p(r; n, \ell) \delta m_{st}^p(r) dr. \quad (8.3)$$

For example, for the density term,

$$S_{mm'}^\rho = -\omega_{n\ell}^2 \mathbf{u}_m^* \cdot \mathbf{u}_{m'}. \quad (8.4)$$

For the λ term,

$$S_{mm'}^\lambda = (\nabla \cdot \mathbf{u}_m)^* (\nabla \cdot \mathbf{u}_{m'}). \quad (8.5)$$

The μ term should be computed using the full tensor contraction

$$S_{mm'}^\mu = 2 \varepsilon(\mathbf{u}_m)^* : \varepsilon(\mathbf{u}_{m'}),$$

and only then projected according to (8.1).

9 Boundary splitting coefficients

The boundary terms can also be projected onto spherical harmonics. For a spherical interface $r = r_\alpha$, define the interface radial/topographic kernel $J_{\alpha s}(n, \ell)$ implicitly by

$$\mathcal{Q}_{mm';st}^{(\ell)} J_{\alpha s}(n, \ell) = r_\alpha^2 \int_{S^2} [\mathcal{B}_{mm'}^\alpha Y_s^t + \mathbf{1}_{\Sigma_\alpha \subset \Sigma_{\text{FS}}} \mathcal{C}_{mm'}^\alpha \cdot \nabla_\Sigma Y_s^t]_{r=r_\alpha} d\Omega. \quad (9.1)$$

Equivalently, for a closed fluid–solid interface one may use the scalar effective boundary kernel:

$$\mathcal{Q}_{mm';st}^{(\ell)} J_{\alpha s}(n, \ell) = r_\alpha^2 \int_{S^2} [\mathcal{B}_{mm'}^\alpha - \nabla_\Sigma \cdot \mathcal{C}_{mm'}^\alpha]_{r=r_\alpha} Y_s^t d\Omega. \quad (9.2)$$

For solid–solid interfaces and the free surface, set $\mathcal{C}^\alpha = 0$.

The topographic contribution to the splitting coefficients is then

$$c_{\text{topo};st}^{(2)}(n, \ell) = \sum_\alpha J_{\alpha s}(n, \ell) \eta_{\alpha;st}. \quad (9.3)$$

This is the corrected replacement for the invalid expression

$$r_\alpha^2 [\lambda_0 K_s^\lambda + \mu_0 K_s^\mu + \rho_0 K_s^\rho].$$

The true $J_{\alpha s}$ is a boundary kernel derived from the differentiated transmission conditions, not a sample of the bulk kernels.

10 Splitting coefficients and splitting function

Combining bulk and topography terms,

$$c_{st}^{(2)}(n, \ell) = \sum_{p \in \{\lambda, \mu, \rho\}} \int_0^a r^2 K_s^p(r; n, \ell) \delta m_{st}^p(r) dr + \sum_\alpha J_{\alpha s}(n, \ell) \eta_{\alpha;st}. \quad (10.1)$$

The splitting matrix is

$$H_{mm'}^{(n,\ell)} = \sum_{s,t} \mathcal{Q}_{mm';st}^{(\ell)} c_{st}^{(2)}(n, \ell). \quad (10.2)$$

The corresponding squared-frequency splitting function is

$$F_{n\ell}^{(2)}(\theta, \phi) = \sum_{s,t} c_{st}^{(2)}(n, \ell) Y_s^t(\theta, \phi). \quad (10.3)$$

If one wants the conventional frequency splitting function, define

$$F_{n\ell}(\theta, \phi) = \frac{1}{2\omega_{n\ell}} F_{n\ell}^{(2)}(\theta, \phi). \quad (10.4)$$

The splitting function is therefore not the Earth model itself. It is a mode-dependent projected observable:

$$\delta m \mapsto \{c_{st}^{(2)}(n, \ell)\}_{s,t} \mapsto H^{(n,\ell)}. \quad (10.5)$$

11 Relation to finite-frequency Fréchet kernels

Liu & Tromp write perturbations to seismic observables in the schematic form

$$\delta\chi = \int_V K_m \delta \ln m \, dV + \int_\Sigma K_d \delta \ln d \, dS + \int_{\Sigma_{\text{FS}}} \mathbf{K}_d \cdot \nabla_\Sigma \delta \ln d \, dS. \quad (11.1)$$

The normal-mode formula derived above is the spectral analogue:

$$\delta\omega^2 = \int_V K_m^{\text{mode}} \delta m \, dV + \int_\Sigma B^{\text{mode}} \eta \, dS + \int_{\Sigma_{\text{FS}}} \mathbf{C}^{\text{mode}} \cdot \nabla_\Sigma \eta \, dS. \quad (11.2)$$

The main difference is that Liu–Tromp kernels are built from a forward wavefield and an adjoint wavefield associated with a chosen misfit measurement, whereas normal-mode splitting kernels are built from a pair of unperturbed eigenfunctions $\mathbf{u}_m, \mathbf{u}_{m'}$ in the degenerate eigenspace.

Thus:

$$\text{finite-frequency wave kernels} \quad \leftrightarrow \quad \text{forward-adjoint wavefield products},$$

whereas

$$\text{normal-mode splitting kernels} \quad \leftrightarrow \quad \text{eigenfunction-eigenfunction bilinear products}.$$

12 Reparameterization to (V_P, V_S, ρ)

In an isotropic solid,

$$V_P^2 = \frac{\lambda + 2\mu}{\rho}, \quad V_S^2 = \frac{\mu}{\rho}. \quad (12.1)$$

Equivalently,

$$\lambda = \rho(V_P^2 - 2V_S^2), \quad \mu = \rho V_S^2. \quad (12.2)$$

For absolute perturbations,

$$\delta\lambda = (V_{P0}^2 - 2V_{S0}^2)\delta\rho + 2\rho_0 V_{P0} \delta V_P - 4\rho_0 V_{S0} \delta V_S, \quad (12.3)$$

$$\delta\mu = V_{S0}^2 \delta\rho + 2\rho_0 V_{S0} \delta V_S. \quad (12.4)$$

Therefore

$$K_s^\lambda \delta\lambda + K_s^\mu \delta\mu + K_s^\rho \delta\rho = K_s^{V_P} \delta V_P + K_s^{V_S} \delta V_S + K_s^{\rho, \text{new}} \delta\rho, \quad (12.5)$$

with

$$\boxed{K_s^{V_P} = 2\rho_0 V_{P0} K_s^\lambda}, \quad (12.6)$$

$$\boxed{K_s^{V_S} = 2\rho_0 V_{S0} (K_s^\mu - 2K_s^\lambda)}, \quad (12.7)$$

$$\boxed{K_s^{\rho, \text{new}} = K_s^\rho + (V_{P0}^2 - 2V_{S0}^2) K_s^\lambda + V_{S0}^2 K_s^\mu}. \quad (12.8)$$

For logarithmic perturbations,

$$\delta V_P = V_{P0} \delta \ln V_P, \quad \delta V_S = V_{S0} \delta \ln V_S, \quad \delta\rho = \rho_0 \delta \ln \rho,$$

so the corresponding log-parameter kernels are

$$K_s^{\ln V_P} = 2\rho_0 V_{P0}^2 K_s^\lambda, \quad (12.9)$$

$$K_s^{\ln V_S} = 2\rho_0 V_{S0}^2 (K_s^\mu - 2K_s^\lambda), \quad (12.10)$$

$$K_s^{\ln \rho} = \rho_0 [K_s^\rho + (V_{P0}^2 - 2V_{S0}^2) K_s^\lambda + V_{S0}^2 K_s^\mu]. \quad (12.11)$$

The same change of variables should not be used to replace the boundary topography kernel by a sampled bulk density kernel. Boundary topography remains governed by $J_{\alpha s}$, computed from (9.1) or (9.2).

13 Inverse problem interpretation

For fixed (s, t) , the splitting coefficients from several modes may be written schematically as

$$c_{st}^{(2)}(n, \ell) = \sum_p \int_0^a r^2 K_s^p(r; n, \ell) \delta m_{st}^p(r) \, dr + \sum_\alpha J_{\alpha s}(n, \ell) \eta_{\alpha; st}. \quad (13.1)$$

Thus, for each angular degree/order (s, t) , one obtains a radial inverse problem:

$$\left\{ c_{st}^{(2)}(n, \ell) \right\}_{n, \ell} = G_s [\{\delta m_{st}^p(r)\}_p, \{\eta_{\alpha; st}\}_\alpha]. \quad (13.2)$$

This shows the central trade-off:

$$\text{bulk structure near an interface} \quad \longleftrightarrow \quad \text{interface topography}.$$

Because $J_{\alpha s}$ is a true boundary kernel, not a sampled bulk kernel, this trade-off cannot be handled correctly by simply evaluating the volumetric kernels at $r = r_\alpha$.

14 Summary

The corrected mathematical pipeline is:

- (1) Define the gravity-free generalized eigenproblem

$$\mathcal{L}_0 \mathbf{u} = \omega^2 \mathbf{M}_0 \mathbf{u}.$$

- (2) For a degenerate multiplet, restrict the first variation of the pencil to

$$\mathcal{E}_{n\ell} = \text{span}\{\mathbf{u}_{n\ell m}\}_{m=-\ell}^\ell.$$

- (3) The bulk splitting matrix is

$$H_{mm'}^{\text{bulk}} = \int_\Omega \left[\delta \lambda S_{mm'}^\lambda + \delta \mu S_{mm'}^\mu + \delta \rho S_{mm'}^\rho \right] \, dV.$$

- (4) The interface-topography part is

$$H_{mm'}^\Sigma = \int_\Sigma \eta \mathcal{B}_{mm'} \, dS + \int_{\Sigma_{\text{FS}}} \mathcal{C}_{mm'} \cdot \nabla_\Sigma \eta \, dS,$$

not a jump of bulk energy density.

- (5) Expanding bulk perturbations and boundary topographies in Y_s^t gives

$$H_{mm'}^{(n, \ell)} = \sum_{s, t} \mathcal{Q}_{mm'; st}^{(\ell)} c_{st}^{(2)}(n, \ell).$$

- (6) The splitting function is

$$F_{n\ell}^{(2)}(\theta, \phi) = \sum_{s, t} c_{st}^{(2)}(n, \ell) Y_s^t(\theta, \phi).$$

A Generalized Hellmann–Feynman identity

Let

$$A(\epsilon)u(\epsilon) = \lambda(\epsilon)B(\epsilon)u(\epsilon),$$

with $A(\epsilon)$ and $B(\epsilon)$ self-adjoint in a fixed Hilbert-space pairing and

$$\langle u(0), B(0)u(0) \rangle = 1.$$

Differentiating at $\epsilon = 0$ gives

$$A'u + Au' = \lambda'Bu + \lambda B'u + \lambda Bu'. \quad (\text{A.1})$$

Taking the inner product with u and using $Au = \lambda Bu$ and self-adjointness yields

$$\boxed{\lambda' = \langle u, A'u \rangle - \lambda \langle u, B'u \rangle.} \quad (\text{A.2})$$

In weak form, this is

$$\boxed{\delta\lambda = \delta a(u, u) - \lambda \delta m(u, u).} \quad (\text{A.3})$$

For a degenerate eigenspace, replace this scalar by the matrix

$$H_{ij} = \delta a(u_i, u_j) - \lambda \delta m(u_i, u_j). \quad (\text{A.4})$$

B Angular identities

The Gaunt integral is

$$\int_{S^2} Y_{\ell_1}^{m_1} Y_{\ell_2}^{m_2} Y_{\ell_3}^{m_3} \, d\Omega = \sqrt{\frac{(2\ell_1 + 1)(2\ell_2 + 1)(2\ell_3 + 1)}{4\pi}} \begin{pmatrix} \ell_1 & \ell_2 & \ell_3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \ell_1 & \ell_2 & \ell_3 \\ m_1 & m_2 & m_3 \end{pmatrix}. \quad (\text{B.1})$$

Since

$$Y_{\ell}^{m*} = (-1)^m Y_{\ell}^{-m},$$

we obtain

$$\int_{S^2} Y_{\ell}^{m*} Y_s^t Y_{\ell'}^{m'} \, d\Omega = (-1)^m \sqrt{\frac{(2\ell + 1)^2 (2s + 1)}{4\pi}} \begin{pmatrix} \ell & s & \ell' \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \ell & s & \ell' \\ -m & t & m' \end{pmatrix}. \quad (\text{B.2})$$

A useful surface-gradient identity is

$$\int_{S^2} (\nabla_h Y_{\ell}^{m*} \cdot \nabla_h Y_{\ell'}^{m'}) Y_s^t \, d\Omega = \frac{1}{2} [2\ell(\ell + 1) - s(s + 1)] \mathcal{Q}_{mm';st}^{(\ell)}. \quad (\text{B.3})$$

This identity is valid after integration over the sphere. It should not be confused with a pointwise identity for the full elastic strain energy density.

C Notes on conventions

Different normal-mode and finite-frequency papers use different normalizations for splitting coefficients. The most common differences are:

- (i) coefficients for $\delta\omega^2$ versus coefficients for $\delta\omega$;
- (ii) absolute perturbations δm versus logarithmic perturbations $\delta \ln m$;

- (iii) topography represented as an absolute normal displacement η versus a relative perturbation $\delta \ln d$;
- (iv) sign conventions for the normal vector $\hat{\mathbf{n}}$ and the jump $\llbracket \cdot \rrbracket$.

In these notes:

$$\llbracket q \rrbracket = q^+ - q^-,$$

$\hat{\mathbf{n}}$ points from the $-$ side to the $+$ side, η is an absolute normal displacement, and $c_{st}^{(2)}$ denotes a coefficient for $\delta\omega^2$.

References

- [1] Dahlen, F. A. and Tromp, J. (1998). *Theoretical Global Seismology*. Princeton University Press.
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